# Information on the Web

## Web Information Architecture, Usability and Accessibility

Spolufinancováno Evropskou unií

MINISTERSTVO ŠKOLSTVÍ, MLÁDEŽE A TĚLOVÝCHOVY

VYSOKÉ UČENÍ TECHNICKÉ V BRNĚ FAKULTA INFORMAČNÍCH TECHNOLOGIÍ

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# Web Information Architecture

# Information Architecture

- **Organization of information**
  - How data is logically organized
- **Labeling (representation) of information**
  - Provides context for the presented information
- **Navigation systems**
  - How users will find their way around the site
- **Search systems**
  - Ways of finding content

# Organizational System

- Organizational **scheme**
  - What criteria we use to categorize information
- Organizational **structure**
  - How we organize the individual categories

# Organizational Scheme

- **Exact schemes** — we must know exactly what we are looking for
  - Alphabetical
  - Chronological
  - Geographical
- **Ambiguous schemes**
  - Topical — article categories, product types, …
  - By activities — what the user wants to achieve
  - By visitor roles — e.g. municipal website: resident, tourist, journalist, …
  - Metaphorical — e.g. organizing the site like a department store (entrance, information desk, departments, …)
  - …or a combination of the above

# Organizational Structures

- **Hierarchical**
  - Easy to understand and remember
  - Easy navigation
- **Database-like**
  - Equivalent documents categorized by metadata
  - E.g. user profile pages
- **General graph**
  - Hypertext links between documents
  - Difficult to navigate

# Labeling Information on the Web

- Headings (labels)
  - Can form a hierarchy
- Contextual links
  - Choice of link text
- Navigation system
  - Labeling items — home page, search, …
- Icons
  - Improve the visual appearance
  - Only a limited number of distinguishable icons

# Important Properties of Labels

- **Specificity**
  - Avoid overly generic labels
- **Consistency**
  - Style — punctuation, capitalisation
  - Presentation — fonts, colours
  - Syntax — naming sections with nouns, questions, etc.
  - Granularity — labelling at the same level of abstraction
  - Completeness — logically related information in the same place
  - Terminology — technical vs. general terms

# Types of Navigation

- Main navigation types
  - Global — which part of the site we are in
    - Always visible
  - Local — what is in this part (e.g. chapter)
    - Varies across sections, should be consistent
  - In-text links
    - Hypertext links to further related information
- Aids
  - Site map
  - Index
  - Guide — linear, step-by-step

# Creating Context

- The user should always know
  - Which website they are on
  - Which section they are in
- A search engine may send a user "into the middle" of a site
- There should always be a clear path
  - To the home page
  - To the beginning of the current section
  - One level up
  - From the root page to the current page (breadcrumb)

# Search

- Not always necessary
- Demanding to implement and operate
  - Or use e.g. Google Programmable Search Engine
- Useful for sites that:
  - Have a large amount of content
  - Have frequently changing content
  - Have an unclear standard navigation
  - Where users expect it
- Search can be over
  - Document content (full-text)
  - Metadata (more demanding, more precise)

# Search System Features

- Interface
  - Search box
  - Advanced search
- Search specification
  - Boolean operators
  - Criteria (restrict which documents are searched)
  - Result presentation options (sorting, detail level)

# Web Usability

# Web Usability

- One of the quality metrics for a website
- Measures the degree of "user comfort" when using the site
- A well-usable website:
  - Visitors can easily find their way around
  - They understand it
  - It does not force users to think — it is intuitive

> *Approximately **40% of visitors** will not return to a site if their initial experience was negative. About **50% of sales are lost** if visitors cannot find what they are looking for fast enough. — Jakob Nielsen and Forrester Research (Failure of Corporate Websites, useit.com, 1998)*

# Importance of Usability

- For commercial websites, usability means **revenue**
  - Poor usability can deter a customer from completing an order
- Properly conducted **usability tests** can reveal obstacles when searching for specific information

> *Studies of user behaviour on the web found a low tolerance for complex designs and slow sites. People **do not want to wait, and do not want to learn** how to use a website. — Jakob Nielsen*

# Usability Factors

- **Learnability**
  - How quickly can a new user complete basic tasks?
- **Efficiency**
  - How efficiently can an experienced user complete assigned tasks?
- **Memorability**
  - Is it easy to remember how to complete a given task?
- **Error tolerance**
  - How easy is it to make a mistake, and how easily can it be corrected?
- **Subjective satisfaction**
  - Do users enjoy using your site?

# Usability vs. Accessibility

- Two often-confused areas
- **Accessibility** — is the content of the site technically available to the user?
- **Usability** — is the user able to make use of the content they can access?

**Accessibility is a necessary but not sufficient condition for usability.**

# Designing a Usable Website

1. Planning
   - What do we want to achieve?
2. User data collection
   - Who will use the site, and what will they expect from it?
3. Prototyping
   - A preliminary version of the site for testing
4. Content collection, writing and revision
   - Selection, organisation and form of content presentation
5. Usability testing
   - Preferably with real users
6. Site promotion and maintenance

# Phase 1: Planning

- Planning helps precisely define the project goals
- First, clarify:
  1. Why we are building the site
  2. Who should use it
  3. When and why they should use it

# Why Are We Building the Site?

- Design **measurable and controllable goals**
- Answer the following questions:
  - How will we know (quantitatively) that the site is successful?
  - What will the consequences be if the site is not successful?
- Avoid overly generic goals
  - E.g. "to inform visitors about our products"
- Better to tie goals to business needs:
  - E.g. "increase mountain bike sales by 30%"
  - "Reduce the load on telephone user support by 50%"

# Who Should Use the Site?

- List the target groups
- Example basic breakdown for an e-commerce site:
  - Users gathering information about products (specs, price, …) — "browsers"
  - Users intending to purchase a specific product — "buyers"
- Depends on the site's focus, e.g. a government institution's site:
  - Citizen, business owner, journalist, …
- Develop a profile for each group:
  - E.g. employed, looking for details, domain-knowledgeable, little web experience
  - These are estimates — they will be verified during the testing phase
- These profiles must be respected in the site's design

# When and Why Users Come

- Users arrive with their own goal in mind
- Create typical scenarios:
  - E.g. Karl's TV has broken down and he needs to buy a new one with similar specs to his old device as quickly as possible.
  - Josef has won the lottery and wants to buy a top-of-the-range TV, but he does not know exactly what specs to look for.
- Further scenarios can be obtained directly from users in later steps

# Phase 2: User Data Collection

- The best way to learn about users is to:
  - **Ask them**
  - **Work with them**
- We want to gather information about:
  - What information they need
  - How they think about and organise information
  - What they expect from the site
  - Their level of domain knowledge
  - Their level of web experience
- We can collect many real usage scenarios

# User Data Collection Techniques

- **Preliminary usability test**
  - Test on the current (old) site, if available, or on a competitor's site
  - Small group of test users (5–12)
  - Observe user behaviour while completing tasks
  - Can also be done remotely (log analysis)
- **Contextual interview**
  - Observe the user in their usual working environment
  - Typically no special tasks are assigned
  - Note environmental characteristics:
    - Technical conditions
    - Workplace atmosphere
    - Can the user ask someone for help?

# User Data Collection Techniques (II)

- **Online surveys**
  - Research using web questionnaires
  - Users fill in information about themselves
  - Important parameters:
    - What information we want to obtain
    - Where we will find respondents
    - The questionnaire should be short (up to 10 questions)
    - Should not be time-consuming to complete
    - Combination of questions about the user (demographics) and their preferences

# User Data Collection Techniques (III)

- **Individual interviews**
  - Smaller number of respondents (around 15)
  - We do not observe behaviour — we gather expectations and preferences
  - Usually a conversation based on a predetermined outline
  - The interview should be conducted by an experienced professional
- **Group discussion (focus group)**
  - Group of 8–12 respondents
  - We gather:
    - User attitudes and requirements
    - Reactions and ideas about prototypes
  - Best led by an experienced moderator

# User Data Collection Techniques (IV)

- **Card sorting**
  - The user participates in the data organisation process
  - The user receives a set of cards with the site's content topics
  - They place the first card on the table
  - For each subsequent card, they decide whether it belongs in the same category or deserves a new one
  - They can rearrange cards at any time
  - The user should "think aloud"

# Phase 3: Prototyping

- A prototype is a "mock-up" of the future site
- Typically does not contain all content
- Contains a sample home page and several pages of different types
- The goal is to have a demonstration of information architecture, navigation and design for user testing
- Allows the design to be verified before investing time and resources into building the full site
- Should be easy to adapt based on findings

# Phase 4: Content Creation

- **Content selection**
  - Is each piece of information relevant to the site's topic?
  - Does the user really need this information?
- **Content organisation**
  - Content must be "easy to use"
  - Written in clear language
  - Presented in an accessible form
  - Logically organised

# Logical Content Organisation

- Break content into manageable chunks
  - Content must not put readers off with its length and complexity
  - Short paragraphs and sentences
  - Lists, tables, images
  - Examples
- Frequent use of headings
  - Simplify orientation on the page
  - Make the page look more interesting
  - Every short section and table should have a heading
- Links to important headings at the top (anchors)
  - For longer documents
  - Help with navigation

# Logical Content Organisation (II)

- Sequential information must be presented in the correct order
  - E.g. understanding one section requires knowledge of another
- If order does not matter, prioritise information that is more important to the user

# Content Presentation

- Use of white space
  - Information should cluster naturally
  - Too much white space feels empty
  - Too little white space — information merges and does not form "clusters"
- Effective expression
  - Every sentence must have meaning — no "filler words"
  - Repeated revisions with time gaps

# Content Presentation (II)

- Language used
  - The site should speak the user's language
  - Avoid specialist "jargon"
- Lists clarify content
  - Bullet points are more comprehensible than long sentences
  - A numbered list is suitable for step-by-step instructions
- Tables
  - More readable than a verbal description
  - E.g. price for individuals, price for companies, …

# Content Presentation (III)

- Examples
  - Users often prefer an example to reading through lengthy theory
- Established presentation forms
  - Many pieces of information are conventionally presented in a specific way
  - E.g. addresses, prices, …
- Icons
  - Liven up the page
  - Emphasise the meaning of words (e.g. phone, …)
- Frequent use of images (where appropriate)

# Users on the Web

- A typical user is in a hurry
  - Looking for specific information
  - Has plenty of other things to do
  - Many pages to browse through
- Web users do not read
  - They only "scan" the content of a document
  - If a document does not look relevant, they look for another
- Reading on a screen can be harder
  - Better to present key information with a link to the full document
  - Provide print-friendly versions of important documents

# Phase 5: Usability Testing

- Testing goals:
  - Can users complete the given task?
  - How quickly?
  - Are they satisfied with that speed?
  - By which paths do they reach the result?
  - Do they find those paths efficient?
  - What are the most frequent problem areas?
  - Do they try approaches the site does not support?

# Phase 6: Promotion and Maintenance

- The site must be findable
  - Placing links on other sites
  - Attractive content ensures more frequent linking
  - Search engine optimisation (SEO)
  - Press releases
  - Regular updates = visitors return more often
- Site maintenance
  - Content updates
  - Link checking
- Future redesign
  - Requirements, technology or users may change

# Web Accessibility

# Accessible Web

- Content should be accessible without requiring specific hardware, software, knowledge or abilities
- Up to **30% of web users** are in some way limited
  - Suboptimal technical equipment
  - Limited display capabilities
  - Health conditions
  - Insufficient experience

# Minority User Groups — Technology

- Users of minority operating systems and browsers
  - Not everyone has Internet Explorer
  - Far fewer have Word or Excel!
- Users of alternative display devices
  - PDAs, smartphones
- Owners of outdated computers
  - Older software versions
  - Low screen resolution
  - Poor or no colour support

# Minority User Groups — Disabilities

- People with dyslexia
  - Do not read poorly structured or long texts
- Colour-blind users (even partially)
  - Struggle to perceive low contrasts
  - E.g. links distinguished only by a colour shade
- Visually impaired users
  - Require the ability to increase text size
- Users with physical disabilities
  - E.g. unable to use a mouse

# Rules for Accessible Web Creation

- International
  - W3C — Web Content Accessibility Guidelines 2.1 (WCAG)
- Czech legislation
  - Act No. 99/2019 Coll. on the accessibility of websites and mobile applications
    - Essentially an application of WCAG 2.1 rules
  - Follows on from Act No. 365/2000 Coll. on public administration information systems — http://www.pravidla-pristupnosti.cz/

# Mandatory Subjects (Czech Law)

The law defines mandatory subjects:

- Territorial self-governing unit
- Legal entity established by law
- Legal entity established or founded by the state or a territorial self-governing unit to satisfy needs in the general interest, financed predominantly from public funds
- A legal entity established by the above
- Voluntary union of municipalities
- University, school and school facility
- Qualified administrator of an electronic identification system

Most of these rules are also well applicable to other websites and will significantly improve their quality.

# WCAG Principles

- **Perceivable**
- **Operable**
- **Understandable**
- **Robust**

# Perceivable

- Provide **text alternatives** for non-text content.
- Provide **captions and alternatives** for audio and video content.
- Create **adaptable content** that can be processed by assistive technologies.
  - Responsive design is required
    - According to WCAG, a page must work at 320 px width (some users zoom in)
  - ARIA attributes (also in previous lecture)
- Ensure **sufficient contrast** between foreground and background so that things are easy to see and hear.

# Operable

- Ensure everything is **accessible from the keyboard**.
- Give users **enough time** to read and interact with page content.
- Do not use content that may cause **seizures**.
- Help users with **navigation and finding** content.
- **Make it easy for users to operate the site with various input devices** (not just the keyboard).

# Understandable

- Write text so that it is **readable and comprehensible**.
- Create content so that it **displays and behaves** as the user expects.
- Help users **avoid and correct mistakes**.

# Robust

- Ensure **maximum compatibility** with current and future technologies, including assistive technologies.

# Best Practice

1. Design the site as accessible from the very beginning
2. Build a skeleton (content structure and navigation) without defining the visual appearance
   - Clean (X)HTML with structural markup and attributes only
   - Maximum use of semantic elements (`header`, `footer`, `section`, …)
3. Add helper links (e.g. skip navigation)
4. Test the usability of navigation without CSS (e.g. with `lynx`)
5. Define the appearance with CSS and hide helper links
   - If the HTML had to be changed, go back to step 4
6. Insert the actual content
7. Throughout the entire process, follow the rules described above

# Auxiliary Content

For users with a screen reader:

- Helper links at the beginning of the page
  - Skip navigation, jump to main content (a reasonable number)
  - `<a href="#content" class="sr-only">Skip to main content</a>`
- Helper headings
  - Label sections that are otherwise distinguished only visually
  - E.g. "Main menu"

# Hiding Content

- **Completely hidden** (sections that should not currently be shown)
  - `display: none;`
  - `visibility: hidden;`
  - Content is not shown; screen readers will not read it

# Hiding Content from Screen Only

For helper links and similar elements that should still be read by assistive technologies:

- Absolute positioning outside the viewport
- Not shown on screen, but screen readers will read it

```html
<a href="#content" class="sr-only">Skip navigation</a>
```

```css
.sr-only {
    position: absolute;
    top: -500px;
    left: 0;
    width: 1px;
    height: 1px;
    overflow: hidden;
}
```

# References


- Web Content Accessibility Guidelines
  https://www.w3.org/TR/WCAG21/
- Accessible web pages — a guide for libraries
  https://prirucky.ipk.nkp.cz/pristupnost/start
- Web Content Accessibility Guidelines (WCAG): an introduction
  https://poslepu.cz/web-content-accessibility-guidelines-wcag-seznamte-se-prosim/
- WAVE Web Accessibility Evaluation Tool
  https://wave.webaim.org/

# That's all! Thank you!